College of Computing and Information Technology

Practicum Student Progress Report

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Name of Group Members (not including the group leader)	Hassan Zahid, Mohammad Sarim Intikhab, Saif Ali, Owais Khalid, Saeed Abdullah Dar	**Progress Report No. (type "1" or "2")**	2
Reporting period (from which week to which?)	Week 8 to Week 11 (up to Saturday 7 November 2026)		
Project Title	Privacy-Preserving Heart Disease Risk Prediction Using Fully Homomorphic Encryption (FHE) and Machine Learning		
Degree Study Program Name of each group member	Naif, Saif: Data and Cyber Security; Sarim, Khalid: Data Science and Artificial Intelligence; Abdullah Dar: Software Engineering; Hassan Zahid: Information Technology		

A. Project Scope

1. Problem statement

Hospitals and clinics increasingly use machine learning to flag patients at risk of heart disease. To make a prediction, the server normally has to see the patient's raw health data, so a breach of that server exposes every patient who used it. Health data is among the most sensitive personal data, and rules such as HIPAA and GDPR limit how it may be shared and stored, which makes many clinics careful about sending it to cloud services. Fully Homomorphic Encryption (FHE) lets a server compute on encrypted data, so in principle it can return a risk prediction without ever seeing the input or the result. The difficulty is cost: FHE only supports addition and multiplication, so functions such as the sigmoid have to be approximated, and encrypted computation is slower and produces much larger data than plaintext. Several of the papers we read do not report a plaintext baseline or the computing costs, so it is hard to tell what this privacy costs for a small, practical model. This project tests, with measured results, whether a heart disease risk model can run on encrypted patient data, and what it costs.

Research questions for the whole project (practicum and Capstone 2):

- RQ-A: Can a logistic regression heart disease risk model run on CKKS-encrypted patient data and give the same low/high-risk decision as the plaintext model for at least 90% of test patients?
- RQ-B: What does the privacy cost, compared with the plaintext model, in accuracy, in time (encryption, inference, decryption), and in ciphertext and key size, on both datasets?
- RQ-C: Can a web system be built in which the patient encrypts in the browser and keeps the private key, so the server only ever handles ciphertext, and which attacks does this design still leave open?

The practicum answers RQ-A and RQ-B on a pilot and reviews the literature (the four SLR review questions in Section A.4); Capstone 2 answers all three on the full prototype.

2. **What is your project's objective?**

General objective: build and evaluate, with measured results, a heart disease risk prediction system in which the server never sees the patient's data or the result in readable form, using FHE. The specific objectives below cover the whole project: the first four are the practicum objectives reported on in this report, and the last four are the Capstone 2 objectives. Each is measurable and independently achievable:

• Practicum: complete a systematic literature review (SLR) of FHE-based machine learning on health data, reported with a PRISMA diagram and a study table that answers four review questions, by Week 12.

• Practicum: train plaintext logistic regression baselines on two datasets (UCI Heart Disease and Edinburgh Myocardial Infarction) and report accuracy, precision, recall, F1 and ROC-AUC on a fixed test split, by Week 12.

• Practicum: replace the sigmoid with a low-degree polynomial and show that its low/high-risk decisions match the baseline on at least 90% of test patients.

• Practicum: run a pilot of CKKS encrypted inference (encrypt, weighted sum plus polynomial, decrypt) with node-seal and TenSEAL, and record encryption time, prediction time, decryption time and ciphertext size, by Week 12. Also produce the architecture and a draft threat model.

• Capstone 2: build the web prototype, with browser-side encryption in node-seal, a Node.js and FastAPI server that computes on ciphertext with TenSEAL, and a result that the patient decrypts locally.

• Capstone 2: secure the API (authentication, input size and format checks, TLS, rate limiting) and complete a security assessment against the threat model drafted in the practicum.

• Capstone 2: benchmark plaintext against FHE on both datasets (accuracy, decision match, encryption, inference and decryption time, ciphertext and key sizes) and deal with the large evaluation keys.

• Capstone 2: deliver the final report and a demo that shows the server only handles ciphertext.

3. **Highlight any changes or adjustments made to the initial goals since the project's start.**

The project goals have not changed. After the feedback on Report 1, the objectives were restated so that they cover the whole project (practicum and Capstone 2) and not only the practicum (Section A.2). The other changes this period are operational: they concern the SLR search strategy and the pilot implementation, not the goals:

• Search strings replaced (29 October). The Week-7 searches used broad strings and returned 1,526 records (1,044 after de-duplication). When screening began in Weeks 8–10, the team found most records irrelevant: the strings did not require all three concepts to appear together, so they matched papers on heart disease and machine learning, or on homomorphic encryption and machine learning, without connecting all three. After the supervisor meeting on 29 October, the team adopted more exact strings that require (homomorphic encryption OR fully homomorphic OR FHE) AND machine learning AND heart disease in the abstract (IEEE Xplore) or in title–abstract–keywords (Scopus). The searches were re-run between 29 October and 6 November: IEEE Xplore returned 8 records and Scopus 34 unique records (35 exported rows, one duplicated record). The PRISMA tracking sheet was rebuilt around this set and the identification counts were restarted.

• IEEE Xplore search string: ("Abstract":"homomorphic encryption" OR "Abstract":"fully homomorphic" OR "Abstract":FHE) AND "Abstract":"machine learning" AND "Abstract":"heart disease".

• Scopus search string: TITLE-ABS-KEY(("homomorphic encryption" OR "fully homomorphic" OR FHE) AND "machine learning" AND "heart disease").

• PubMed was not re-run with the refined string. PubMed was added in Report 1 only as a replacement for the ACM Digital Library; with the refined strategy the team searches the two databases that index the computing literature

where FHE-based machine learning work is published (IEEE Xplore and Scopus). Scopus also covers approximately 6,700 of the roughly 7,000 MEDLINE journal titles [9], so the biomedical literature that PubMed would return is already represented inside the Scopus search. This is recorded as a limitation of the review.

Justification: screening a mostly irrelevant set of 1,044 records would consume the remaining term for little gain, while the refined set of 34 records is much more focused on the papers the four review questions target. The change followed the supervisor's advice and is documented in the PRISMA tracking sheet.

• Pilot client side (Weeks 10–11): Report 1 planned patient-side encryption in the browser with node-seal. The cost measurements were first taken with the client side in TenSEAL itself (the “encrypt in a local Python client” option of Plan B in Report 1), and the full cross-library round-trip was then completed on a real test patient: node-seal 7.0.0 encrypted the 13 features, TenSEAL 0.3.16 computed the encrypted score, and node-seal decrypted it – the decrypted score matched the plain model exactly (see Table 5). One environment finding: node-seal 7.0.0 requires Node.js 22 or newer (its WebAssembly build uses the GC and exception-handling proposals); Node.js 20 cannot load it.

• Ciphertext layout: the pilot measured both layouts. The packed layout (all 13 features in one ciphertext, dot product) needs only 1 MB per patient but requires 164 MB of Galois (rotation) keys; the one-ciphertext-per-feature layout planned in Report 1 (used in the cross-library round-trip) needs no rotation keys at all but about 16 MB of ciphertexts per patient. The final choice is deferred to Capstone 2 (see Section C for the mitigations).

4. Methodology / Approach / Workflow

Part 1: Systematic literature review. We follow Kitchenham and Charters [2] and report with PRISMA 2020 [1]. The protocol keeps the four review questions from Report 1: (RQ1) which FHE schemes and libraries are used for ML on health data; (RQ2) which models and sigmoid approximations are used and how encrypted accuracy compares with plaintext; (RQ3) what computing costs are reported; (RQ4) how keys and client-side encryption are handled. Databases: IEEE Xplore and Scopus, 2016–2026, English, journal and conference papers, using the refined strings above. Each search was run by one member and re-run by a second member to confirm the count. The 34 records are divided among the six members, and each member screens the records assigned to them in the PRISMA tracking sheet (Excel) against the inclusion and exclusion criteria, logging the decision, the reason and whether only the title and abstract or the full paper was checked. Borderline records are discussed by the whole team and a second member re-checks the excluded records. Records that stay in go on to full-text reading, a quality score out of 5 and data extraction. Review-type records are excluded but kept for snowballing their reference lists. Records whose full text we cannot yet access stay undecided until access is arranged and are excluded if it cannot be.

Part 2: Data and baseline models. Two public datasets (data statement): UCI Heart Disease, Cleveland [7], and Edinburgh Myocardial Infarction [8]. For each, we clean and scale the data, use one fixed stratified train/test split, and train logistic regression with scikit-learn; the same split is later used to compare the encrypted and plaintext models.

Part 3: Encrypted inference pilot. The CKKS scheme [3] is used because it handles real-valued features such as blood pressure. The server uses TenSEAL 0.3.16 [4] and the patient-side browser uses node-seal 7.0.0 [5]; both build on Microsoft SEAL [6]. Because FHE only supports addition and multiplication, the sigmoid is replaced by a low-degree polynomial. The pilot measures encryption, inference and decryption times, and ciphertext and key sizes; all 13 features of a patient are packed into one ciphertext and the weighted sum is computed as a dot product (see B.2 for the trade-off against the per-feature plan in Report 1). Library versions are pinned: in a compatibility test, ciphertexts from node-seal 7.0.0 (SEAL 4.1.2) were read and computed on correctly by TenSEAL 0.3.15 and 0.3.16, while TenSEAL 0.3.17 and 0.3.18 reported an incompatible version.

Part 4: Capstone 2 (planned). The pilot is extended into a working prototype with an HTML/JavaScript front end where the patient enters the 13 features and encrypts them in the browser with node-seal, a Node.js (Express) server that only

routes requests, and a Python (FastAPI) service that runs the encrypted dot product and polynomial sigmoid with TenSEAL and returns the encrypted score, which only the patient can decrypt. Work follows this order: front end, API and plaintext baseline app; the real FHE pipeline; security testing of the API against the threat model; performance evaluation (plaintext against FHE on the same test split of both datasets, measuring accuracy, decision match, times, and ciphertext and key sizes); then polish and the final demo. The ciphertext layout (packed or one per feature) is chosen at the start of Capstone 2 from the pilot measurements in Table 5.

Tools: GitHub, Python, scikit-learn, TenSEAL, node-seal, draw.io, Excel (PRISMA sheet). All work is in the private repository: https://github.com/Saeed-60307149/fhe-heart-risk

5. Data Statement

Both project datasets are existing, de-identified, public tabular datasets: UCI Heart Disease (Cleveland) [7] (303 samples, 13 features + label) and Edinburgh Myocardial Infarction [8] (1,253 samples, 9 features + label). We collect no data from people and involve no human participants; the demo uses only test-set rows. Because the datasets have different variables, results are reported separately, not pooled. For the review, records (titles and abstracts) were exported from IEEE Xplore and Scopus via UDST Library access and are kept in the private repository for review use only. No IRB approval is required: both datasets are public and fully de-identified, and no data are collected from people.

B. Project progress

This period concentrated on correcting the SLR search strategy and on the encrypted inference pilot. Progress slowed in Weeks 8 and 9: Week 8 was dominated by mid-term examinations and Week 9 was the semester break, so screening of the Week-7 record set stalled after only a small number of records had been checked – enough, however, to show that the broad strings produced mostly irrelevant records. Screening resumed in Week 10. Following the supervisor meeting on 29 October, the team defined more exact strings, re-ran both database searches and rebuilt the PRISMA tracking sheet: 43 records identified (IEEE Xplore 8; Scopus 35 rows), 9 duplicates removed, leaving 34 unique records ready for screening. These were screened between 1 and 5 November; the outcome is in Tables 2 and 3. In parallel, the encrypted inference pilot – run in TenSEAL 0.3.16 as a local Python client (Plan B of Report 1; see B.2) – ran end-to-end on the UCI Cleveland data: the same logistic regression model executed on CKKS-encrypted features with no loss of accuracy, and the full cost measurements (encryption, inference and decryption times; ciphertext and key sizes) are reported in Table 5. The full cross-library round-trip from Report 1's plan was then completed on a real test patient: node-seal 7.0.0 encrypted the 13 features in the client, TenSEAL computed the encrypted score on the server, and node-seal decrypted the result – the decrypted score matched the plain model exactly (0.877).

PRISMA identification stage (refined searches, 29 Oct – 6 Nov 2026)	**Records**	**Run / confirmed by**
IEEE Xplore (abstract; 2016–2026; conferences and journals)	8	Run by Naif; re-run and confirmed by Saif
Scopus (title, abstract, keywords; 2016–2026; English)	35 rows (34 unique)	Run by Naif; re-run and confirmed by Abdullah
Total records identified	43	Run by Naif; re-run and confirmed by Abdullah
Duplicates removed (8 same-DOI across databases + 1 within Scopus)	9	Run by Naif; re-run and confirmed by Saif
Records to screen	34	everyone

Table 1: Records found and removed after search refinement.

Screening: the 34 records were divided among the six members (five or six each) and screened between 1 and 5 November. Three records appeared twice in the sheet (Heartify, Sayyad and Kulkarni, and the ICSCSP 2018 proceedings entry) and were read by two members each, so 31 unique records were screened. Eleven were excluded: 3 review articles (kept for snowballing), 3 conference-proceedings entries, 1 retracted article, 1 with no health data, 1 with no encryption, 1 federated-learning paper that does not use HE, and 1 Paillier-only chapter with no results. Twelve are included and go on to full-text quality scoring and extraction, and 2 are borderline

(both use Paillier, which only adds) and will be decided on the full text. Six records are waiting for their full text through UDST access and stay undecided until we get it; if access is not possible they will be excluded and reported as a limitation. Figure 1 shows the flow.

Screening step (up to 7 Nov 2026)	Records
Records to screen (after de-duplication, Table 1)	34
Rows listed twice under the same title (counted once)	3
Unique records screened	31
Excluded (reasons in Table 3)	11
Included, going on to full-text scoring and extraction	12
Borderline, to be decided on the full text (MediVault, Akter)	2
Full text not yet accessible, awaiting access	6

Table 2: Screening outcome, 1 to 5 November 2026.

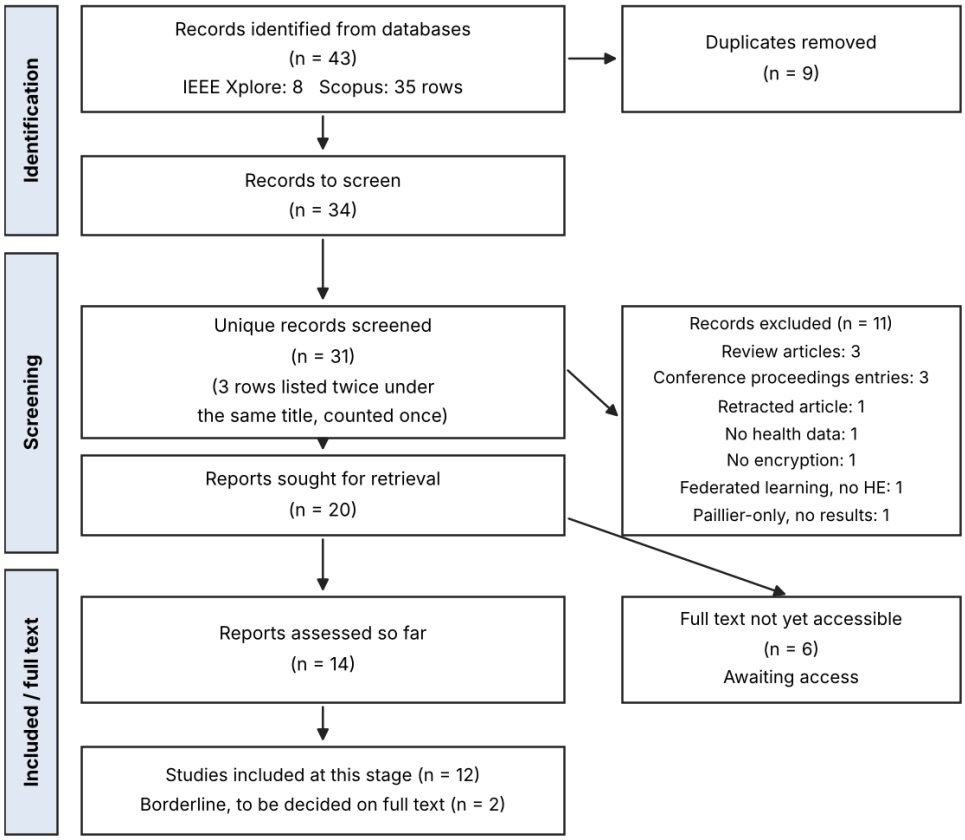


Figure 1: PRISMA flow diagram of the screening, up to 7 November 2026.

Table 3 lists every record, who screened it, whether only the title or abstract, the full text, or just the database record was checked, and the decision with its reason.

#	Record	Reviewer	Checked	Decision	Reason
1	AutoPro-RHC, Sheu et al., 2023	Saeed	Full text	Include	TFHE logistic regression and SVC on UCI heart data through Concrete ML; accuracy reported.
2	TrustFedHealth, Dang et al., 2023	Saeed	Abstract	Include	Federated learning with HE on heart (ECG) data; passed the first stage.
3	XAI with HE for cloud ECG, Cenitta et al., 2025	Saeed	Full text	Include	CKKS with a CNN on encrypted ECG; 94.2% plain vs 93.4% encrypted, about 420 ms per prediction.
4	Data mining and ML healthcare monitoring in Cloud-IoT	Hassan	Full text	Include	HE on the Cleveland heart data; 90% accuracy reported.
5	Privacy-preserving cloud data mining with HE-based logistic regression	Hassan	Full text	Include	Encrypted logistic regression on UCI Heart Disease; 84.2% accuracy.

#	Record	Reviewer	Checked	Decision	Reason
6	Secure ML evaluation with FHE and Concrete ML	Hassan	Full text	Include	FHE inference on heart disease data with LR, SVM, trees and XGBoost.
7	Logistic regression training with approximate HE, Kim et al.	Saif	Full text	Include	Encrypted logistic regression with HEAAN; times and memory reported, Edinburgh dataset used.
8	FHE-driven logistic regression for heart disease, Naresh and Reddi, 2025	Saif	Full text	Include	CKKS with TenSEAL; 84% encrypted vs 85% plain.
9	ANN activation function estimators for HE inference, 2025	Naif	Full text	Include	Activation approximations under HE with a heart disease example; relevant to RQ2.
10	Outsourced privacy-preserving decision tree classification, Wang et al., 2020	Naif	Full text	Include	Encrypted tree classification; heart disease tree timings reported.
11	GuardML, Frimpong et al., 2024	Khalid	Full text	Include	Hybrid HE (PASTA with BFV) on ECG data; 87.55% encrypted accuracy.
12	Nguyen et al., 2024 (IEEE DASC)	Khalid	Full text (partial)	Include	Same authors, scheme and dataset as GuardML, so counted as overlapping evidence.
13	MediVault, Li et al., 2026	Saif	Full text	Borderline	Federated learning with Paillier (addition only) on heart data; depends on whether partial HE counts.
14	Blockchain, k-medoids and HE, Akter et al., 2022	Naif	Full text	Borderline	Paillier (addition only) on heart and other data; same question as MediVault.
15	Privacy preserving inference over encrypted data, Sayyad and Kulkarni, 2022	Khalid, Sarim	Abstract	Awaiting access	HElib on UCI Heart Disease; full text not accessible yet, no results in the abstract.
16	Secure decision tree evaluation with FHE, Sun et al., LNCS 16176	Naif	Abstract	Awaiting access	Full text behind a paywall; awaiting access. Abstract: FHE decision tree evaluation, health data use not confirmed.
17	HE and KNN for heart disease, Behera et al., 2022	Sarim	Partial page	Awaiting access	Full text not accessible yet; page mentions both partial and full HE.
18	Federated learning frameworks for heart disease, Kapila et al., 2025	Sarim	Abstract	Awaiting access	Full text not accessible yet; role of HE unclear from the abstract, likely related work.
19	Yadkar et al., 2025 (IEEE ICBATS)	Khalid	Record only	Awaiting access	Full text not accessible yet.
20	Sravani and Kumar, 2025 (CRC Press)	Khalid	Record only	Awaiting access	Full text not accessible yet.
21	Heartify, Sinciya et al., 2025	Hassan, Saeed	Full text	Exclude	Federated learning only, no homomorphic encryption used.
22	Privacy-preserving federated learning for heart disease prediction: a review	Hassan	Full text	Exclude	Review article with no experiments of its own; kept for snowballing.
23	Review of explainable federated multimodal AI for heart disease, Murali and Meenatchi, 2026	Saif	Full text	Exclude	Review article; proposed framework never built; kept for snowballing.
24	Secure healthcare monitoring survey, Poonam and Mallapur, 2025	Saif	Full text	Exclude	Short survey; kept for snowballing.
25	Blockchain, HE and voting classifier in IIoT, Nisha et al., 2023	Saeed	Record only	Exclude	No health or patient data found; abstract not found.
26	IC-ICN 2020 conference review	Saeed	Record only	Exclude	Proceedings entry, not a study.
27	ICSCSP 2018 conference review	Saif, Sarim	Record only	Exclude	Proceedings entry, not a study.
28	ICDSIS 2022 conference review	Naif	Record only	Exclude	Proceedings entry for 134 papers, not a study.
29	Naresh and Varma, 2025 (Smart Health)	Khalid	Full text	Exclude	Retracted article (stroke data).
30	CNN and random forest for cardiovascular disease, Shruthi and Naidu, 2025	Sarim	Record only	Exclude	No encryption in the title; abstract and full text not found.
31	Secure healthcare predictive analysis with HE, Gogoi and Valan, 2025	Sarim	Abstract	Exclude	Paillier case study, no results in the abstract, full text paywalled; kept as background reading.

Table 3: Screening decision for each of the 31 unique records (HE = homomorphic encryption).

1. Ongoing tasks

• Getting the full text of the 6 records waiting for access and settling the 2 borderline records (Week 12), then a second-reviewer re-check of the excluded records.

• Full-text screening, quality scoring and data extraction for the included records.

• Refining the polynomial sigmoid approximation (the pilot already uses a least-squares cubic, mean absolute error 0.03 on the test set) and validating the 95% decision match against the plaintext baseline.

• Extending the encrypted pilot to the Edinburgh Myocardial Infarction dataset (Week 12).

• Threat-model drafting and architecture diagram finalisation.

2. Completed tasks

Task	Who	When
Screening of the Week-7 (broad-string) record set started (first records showed low relevance)	All (pairs)	Week 8
Screening paused: mid-term examinations (Week 8) and semester break (Week 9)	All	Weeks 8–9
Supervisor meeting: SLR direction and more exact search strings	All	29 Oct
Refined search strings defined and validated	Naif, Saif	29 Oct
Searches re-run and exported: IEEE Xplore (8 records), Scopus (35 rows)	Naif	29 Oct – 6 Nov
Search counts re-run and confirmed: IEEE Xplore (Saif), Scopus (Abdullah)	Saif, Abdullah	29 Oct – 6 Nov
PRISMA tracking sheet rebuilt: 43 identified, 9 duplicates removed, 34 to screen	Naif	6–7 Nov
Title/abstract screening of the 34 refined records (31 unique); every decision logged in Table 3	All (six members)	Weeks 10–11 (1–5 Nov)
Encrypted pilot environment set up with the pinned versions (node-seal 7.0.0, TenSEAL 0.3.16)	Saif	Weeks 8–10
Encrypted inference pilot completed on UCI Cleveland: model trained, encrypted round-trip executed, cost measurements taken (Table 5)	Saif, Naif	Weeks 10–11
Cross-library round-trip on a real patient: node-seal encrypt, TenSEAL compute, node-seal decrypt (score matches the plain model exactly)	Saif, Naif	Week 11
Technical facts for the threat model compiled	Naif	Week 10

Table 4: Completed tasks in the reporting period.

3. Upcoming tasks

• Get the 6 waiting full texts and settle the 2 borderline records (Week 12) and record the final PRISMA numbers.

• Full-text screening, quality scoring and data extraction; snowball the reference lists of the excluded review papers (Weeks 12–13).

• Finalise the plaintext baselines on both datasets (Weeks 11–12).

• Finalise the polynomial sigmoid and validate the 95% decision match (Weeks 12–13).

• Extend the pilot cost measurements to the Edinburgh dataset; finalise the architecture and threat model (Weeks 12–13).

• Final report and presentation preparation (Weeks 14–15).

4. Project work distribution strategy

The team works in three degree-aligned pairs, as agreed in Progress Report 1: Pair 1 (Naif and Saif, Data and Cyber Security) runs the searches (Naif ran them, Saif re-ran the IEEE Xplore one) and the encrypted pilot; Pair 2 (Sarim and Hassan Zahid) leads the datasets, de-duplication and PRISMA record tracking; Pair 3 (Owais Khalid and Saeed Abdullah Dar) re-ran the Scopus search (Abdullah) and keeps the repository and the protocol documentation. Each PRISMA record is screened by its assigned reviewer, and each member logs at least six hours per week. When the SLR stalled in Weeks 8–10, tasks were redistributed temporarily (Saif took the encrypted pilot so Naif could focus on the search strategy). For the current screening round, the leader divided the 34 refined records across the six team members, five to six records each, and the whole sheet was screened before the Progress Report 2 deadline (Sunday 8 November); borderline verdicts are discussed with the whole team.

5. Gantt chart / Timeline

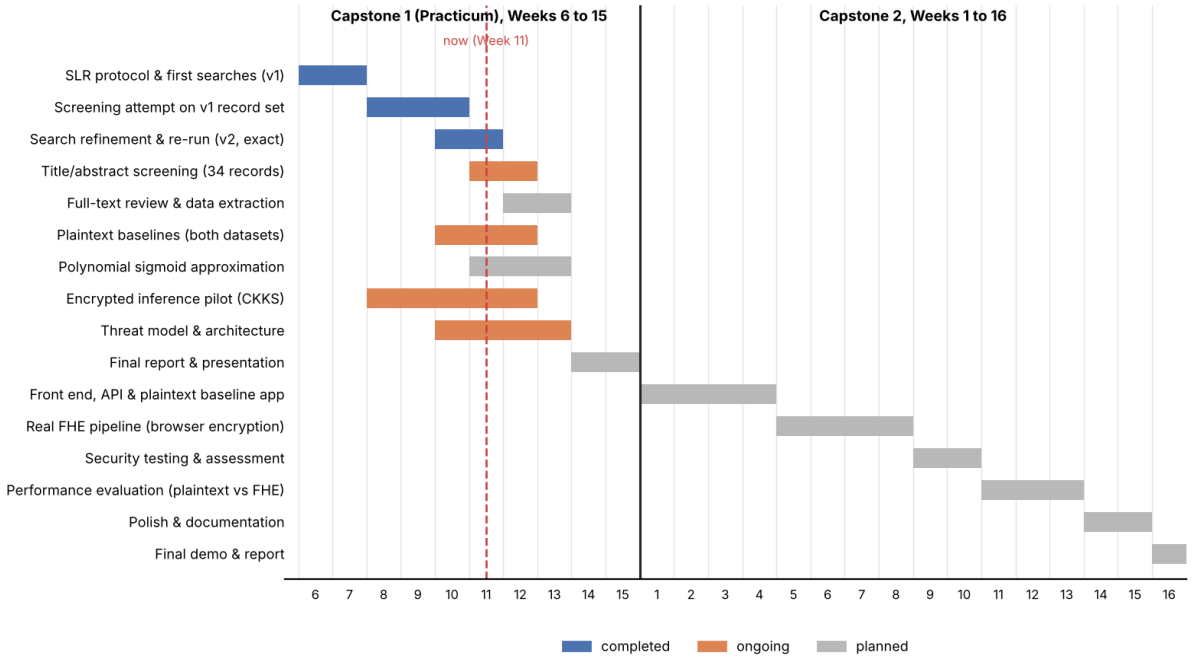


Figure 2: Updated Gantt chart from Capstone 1 (Practicum, Weeks 6 to 15) through Capstone 2 (Weeks 1 to 16).

C. Challenges Encountered

The main challenge this period was methodological (search precision) rather than technical; it is described below together with the mitigations. Challenges are raised at the weekly supervisor meeting before they are written up here: the search problem was discussed with the supervisor on 29 October, and every new challenge is put on the agenda of the next weekly meeting.

1. General challenges (technical/financial/others in relation to the project)

• Low search precision (Weeks 8–10): the broad Week-7 strings returned 1,044 records after de-duplication, but most were irrelevant to FHE-based heart disease prediction, so screening effort produced few usable papers. Mitigation: supervisor meeting on 29 October; new exact strings requiring all three concepts; searches re-run and the PRISMA sheet rebuilt around 34 records.

• Library version incompatibility: TenSEAL 0.3.17/0.3.18 reject node-seal 7.0.0 ciphertexts, and node-seal 7.0.0 itself requires Node.js 22+ or a current browser (its WebAssembly build uses the GC and exception-handling proposals; Node.js 20 cannot load it). Mitigation: versions pinned (node-seal 7.0.0, TenSEAL 0.3.16) and the development environment upgraded to Node.js 22; Plan B documented (run node-seal on a Node.js server, or encrypt in a local Python client).

• Large evaluation keys: the packed one-ciphertext-per-patient layout used in the pilot (instead of the per-feature plan in Report 1) needs Galois (rotation) keys, and the polynomial degree required for the cubic sigmoid (16,384) makes them 164 MB – impractical to ship to a browser client. Mitigation planned for Capstone 2: SEAL key compression, a lower-degree approximation, or a return to one ciphertext per feature, which avoids rotation keys entirely.

• Retracted article in the results: one Scopus record (heart-stroke detection with federated learning and homomorphic encryption, Smart Health) is retracted. Mitigation: excluded from the review and flagged in the PRISMA sheet.

• Capstone 2 risk: the pilot showed encrypted inference is fast (about 88 ms per patient) but the 164 MB evaluation keys and browser support for node-seal limit the web design. Mitigation: the ciphertext layout and key size are the first Capstone 2 design decision, using Table 5.

2. Group teamwork/leadership challenges

The SLR setback in Week 10 put the review pair behind schedule. The team redistributed the work for that week (Saif took over the encrypted pilot while Naif worked on the search strategy) and used the weekly meeting to keep everyone aligned.

3. How challenges were resolved/mitigation plan

• Weekly team meetings and supervisor checkpoints; all protocol changes are logged in the PRISMA sheet.

• Screening verdicts are recorded in the shared PRISMA sheet with reason codes, and borderline cases are discussed by the whole team, and a second reviewer will re-check the excluded records, so a single reviewer's error does not decide a record's fate.

• Version pinning and a documented Plan B for the FHE libraries.

• Records are divided among the members so the screening load is shared, and the excluded records are cross-checked by a second reviewer.

D. Expected Results and/or Findings (and optional pilot results)

Linked to the objectives, the project expects the following results. The first four are for the end of the practicum and the last four for the end of Capstone 2:

• Practicum: a completed SLR reported with a PRISMA 2020 flow diagram (identification, screening, included) and a study table answering RQ1–RQ4, measured by the final PRISMA counts and a completed extraction table.

• Practicum: plaintext logistic regression baselines on both datasets with accuracy, precision, recall, F1 and ROC-AUC on a fixed split.

• Practicum: a low-degree polynomial sigmoid whose risk decisions match the baseline on at least 95% of test patients.

• Practicum: a working CKKS encrypted round-trip pilot with measured encryption, inference and decryption times and ciphertext and key sizes, plus the system architecture and threat model.

• Capstone 2: a working web prototype in which the patient encrypts in the browser and the server only ever handles ciphertext, shown in a demo that includes the ciphertext visible in the browser's developer tools.

• Capstone 2: a secured API and a security assessment of the whole system against the threat model.

• Capstone 2: a benchmark report comparing plaintext and FHE on both datasets (accuracy, decision match, times, ciphertext and key sizes), answering RQ-A and RQ-B in full.

• Capstone 2: the final report and presentation.

Contribution: together, the review and the pilot provide evidence on whether private, server-side heart disease risk prediction with FHE is practical for clinics, and at what cost in accuracy, speed and data size. The pinned node-seal/TenSEAL workflow is a reusable recipe for browser-side encrypted inference.

Pilot measurement (CKKS, degree 16,384, scale 2^{40})	**Value**
Key generation time (one-off)	1.18 s
Encryption / inference / decryption time per patient	13.3 / 87.5 / 3.5 ms
Ciphertext size (13 features, 1 patient)	1,029 KB
Key sizes: public context / with secret key / Galois keys	7.7 MB / 1.9 MB / 163.9 MB
Plain-model accuracy (75-patient test set)	0.853 (ROC-AUC 0.947)
Encrypted-inference accuracy	0.853 (no loss vs plain model)
Sigmoid approximation error (mean absolute)	0.031
node-seal encryption (browser side, per feature, Node.js 22)	33.4 ms/feature (434 ms per patient)
node-seal decryption (client side)	3.3 ms
Ciphertext size, per-feature layout (13 ciphertexts)	about 16 MB per patient

Table 5: Encrypted inference pilot measurements, UCI Cleveland dataset (297 patients, 75-patient test split, TenSEAL 0.3.16). Figures from the pilot run of 7 November 2026; re-run and confirmed by the team before submission (the re-run was done by Saif). Edinburgh MI measurements are scheduled for Week 12.

E. Conclusion

In this period the project corrected its main methodological weakness, the SLR search strategy, and delivered its first end-to-end encrypted result. The refined, exact strings replaced a mostly irrelevant set of 1,044 records with 34 records; 31 unique records have now been screened, 11 excluded, 14 kept for full-text assessment (12 included, 2 borderline) and 6 waiting for full-text access. The encrypted inference pilot ran the full logistic regression model on CKKS-encrypted UCI Cleveland features with no loss of accuracy (0.853 on both plain and encrypted paths, ROC-AUC 0.947), at a cost of about 13 ms encryption, 88 ms encrypted inference and 4 ms decryption per patient, with a 1 MB ciphertext per patient; the cross-library round-trip (node-seal encryption, TenSEAL computation, node-seal decryption) reproduced the plain model's score exactly on a real patient – evidence that privacy-preserving heart disease risk prediction is practical for this workload. The main open items are completing the screening, extending the pilot to the Edinburgh dataset and shrinking the 164 MB evaluation keys; with these scheduled for Weeks 12–13, the project remains on track to meet its practicum objectives by the end of the term, and Capstone 2 will turn the pilot into the web prototype, secure it and benchmark it (Sections A.2, A.4 and D).

F. Acknowledgements

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G. Conflict of Interest Statement (Optional)

The authors declare no conflict of interest.

H. References

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